

Benchmark results compiled with -O2: mean time (microseconds)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective-naive	mp	mp.safe	mtl	handrolled	mtl.listt	mtl-logict	handrolled.listt	handrolled.logict
countdown.shallow.10000	166.255	157.239	1256.015	4.729	150.896	295.271	265.388	89.929	197.401	1067.645	4.816	4.76	–	–	–	–
countdown.deep.10000	1289.38	581.548	2330.969	2818.884	150.442	286.934	167.87	171.407	200.406	3584.87	4.898	4.835	–	–	–	–
catch.shallow.10000	283.116	36.006	1163.213	21.246	1077.618	379327	252.051	74.686	859.381	853.846	20.919	20.944	–	–	–	–
catch.deep.10000	3262.75	33.911	6723.847	1712.242	1090.063	397005.9	180.75	244.954	866.516	863.507	21.079	21.093	–	–	–	–
local.shallow.10000	955.407	467.497	2206.931	4.966	554.016	406894.4	203.759	146.694	799.416	801.309	5.048	4.998	–	–	–	–
local.deep.10000	3881.405	466.802	5192.7	5.013	553.463	392608.8	208.085	219.046	807.983	814.283	5.067	5.006	–	–	–	–
nondet.shallow.32	2000.769	3364.342	7949.819	263.806	–	25124.037	794.52	920.823	1799.283	1790.316	–	–	389.365	227.925	377.133	250.615
nondet.deep.32	5484.113	5829.288	8050.619	2509.983	–	51205.05	776.646	641.276	5838.881	5809.777	–	–	63635.15	247.734	65396.425	248.312

Benchmark results compiled with -O2: mean time (% of fastest)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective-naive	mp	mp.safe	mtl	handrolled	mtl.listt	mtl-logict	handrolled.listt	handrolled.logict
countdown.shallow.10000	3515.9%	3325.3%	26561.9%	100.0%	3191.1%	6244.3%	5612.4%	1901.8%	4174.6%	22578.3%	101.8%	100.7%	–	–	–	–
countdown.deep.10000	26670.2%	12029.0%	48214.9%	58307.2%	3111.8%	5935.1%	3472.3%	3545.5%	4145.3%	74151.2%	101.3%	100.0%	–	–	–	–
catch.shallow.10000	1353.4%	172.1%	5560.5%	101.6%	5151.4%	1813306.4%	1204.9%	357.0%	4108.1%	4081.7%	100.0%	100.1%	–	–	–	–
catch.deep.10000	15478.7%	160.9%	31898.3%	8123.0%	5171.3%	1883419.8%	857.5%	1162.1%	4110.8%	4096.5%	100.0%	100.1%	–	–	–	–
local.shallow.10000	19240.9%	9414.9%	44445.2%	100.0%	11157.3%	8194414.7%	4103.5%	2954.3%	16099.4%	16137.5%	101.7%	100.7%	–	–	–	–
local.deep.10000	77542.6%	9325.8%	103739.6%	100.1%	11057.1%	7843526.2%	4157.1%	4376.1%	16141.9%	16267.7%	101.2%	100.0%	–	–	–	–
nondet.shallow.32	877.8%	1476.1%	3487.9%	115.7%	–	11023.0%	348.6%	404.0%	789.4%	785.5%	–	–	170.8%	100.0%	165.5%	110.0%
nondet.deep.32	2213.7%	2353.0%	3249.7%	1013.2%	–	20669.4%	313.5%	258.9%	2356.9%	2345.2%	–	–	25686.9%	100.0%	26397.8%	100.2%

Benchmark results compiled with -O2: allocated bytes

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective-naive	mp	mp.safe	mtl	handrolled	mtl.listt	mtl-logict	handrolled.listt	handrolled.logict
countdown.shallow.10000	1760143	1600142	13361463	24	1121559	2087372	2480238	1040119	3441166	14241159	24	24	–	–	–	–
countdown.deep.10000	17760453	10400192	37363454	27121221	1136503	2088075	2000190	3920263	3442679	41443884	24	24	–	–	–	–
catch.shallow.10000	4069603	65595	7764934	65585	7846335	413419360	1687790	723947	6081375	6081376	65585	65585	–	–	–	–
catch.deep.10000	16870399	66057	55153607	16165070	7864080	414232192	1527781	3367869	6083937	6083936	65585	65585	–	–	–	–
local.shallow.10000	7205040	4793283	12630003	24	4326240	409640173	1669445	1029429	5513532	5513563	24	24	–	–	–	–
local.deep.10000	16005982	4793724	30232155	24	4341323	410400285	1589469	2549549	5515779	5515780	64	64	–	–	–	–
nondet.shallow.32	13454088	1588979	43492459	3606902	–	111260645	1531312	1532745	22646039	22646045	–	–	3664100	1987325	3664078	2003981
nondet.deep.32	67560245	45645983	43569828	26810657	–	383225914	1531392	1535143	68636399	68636166	–	–	469818607	1969093	469818389	2010669

Benchmark results compiled with -O2: allocated bytes (% of least allocated)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective-naive	mp	mp.safe	mtl	handrolled	mtl.listt	mtl-logict	handrolled.listt	handrolled.logict
countdown.shallow.10000	7333929.2%	6667258.3%	55672762.5%	100.0%	4673162.5%	8697383.3%	10334325.0%	4333829.2%	14338191.7%	59338162.5%	100.0%	100.0%	–	–	–	–
countdown.deep.10000	74001887.5%	43334133.3%	155681058.3%	113005087.5%	4735429.2%	8700312.5%	8334125.0%	16334429.2%	14344495.8%	172682850.0%	100.0%	100.0%	–	–	–	–
catch.shallow.10000	6205.1%	100.0%	11839.5%	100.0%	11963.6%	630356.6%	2573.4%	1103.8%	9272.5%	9272.5%	100.0%	100.0%	–	–	–	–
catch.deep.10000	25723.0%	100.7%	84094.8%	24647.5%	11990.7%	631595.9%	2329.5%	5135.1%	9276.4%	9276.4%	100.0%	100.0%	–	–	–	–
local.shallow.10000	30021000.0%	19972012.5%	52625012.5%	100.0%	18026000.0%	1706834054.2%	6956020.8%	4289287.5%	22973050.0%	22973179.2%	100.0%	100.0%	–	–	–	–
local.deep.10000	66691591.7%	19973850.0%	125967312.5%	100.0%	18088845.8%	171080187.5%	6622787.5%	10623120.8%	22982412.5%	22982416.7%	266.7%	266.7%	–	–	–	–
nondet.shallow.32	878.6%	1037.7%	2840.2%	235.5%	–	7265.7%	100.0%	100.1%	1478.9%	1478.9%	–	–	239.3%	129.8%	239.3%	130.9%
nondet.deep.32	4411.7%	2980.7%	2845.1%	1750.7%	–	25024.7%	100.0%	100.2%	4482.0%	4481.9%	–	–	30679.2%	128.6%	30679.2%	131.3%

Benchmark results compiled with -O0: mean time (microseconds)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective-naive	mp	mp.safe	mtl	handrolled	mtl.listt	mtl-logict	handrolled.listt	handrolled.logict
countdown.shallow.10000	813.321	520.988	1773.492	1729.885	730.394	888.71	1632.845	1668.526	864.224	2269.838	671.374	572.821	–	–	–	–
countdown.deep.10000	2985.027	994.746	2875.936	10917.1	746.676	890.843	1699.94	3343.78	1469.318	5295.312	5220.016	6010.422	–	–	–	–
catch.shallow.10000	1094.155	308.035	2820.794	3420.3	2148.606	378281.4	944.095	1071.595	1643.405	1647.855	259.155	226.863	–	–	–	–
catch.deep.10000	8193.331	318.993	7077.137	16050.263	2165.831	393021	1053.085	3002.828	1660.324	1657.971	4006.105	4084.575	–	–	–	–
local.shallow.10000	4515.694	672.989	2706.161	1451.068	1073.754	387670.1	642.772	661.102	1423.03	1417.495	358.752	336.373	–	–	–	–
local.deep.10000	13477.7	686.443	5642.959	6065.353	1082.858	407835.6	628.914	1631.923	1401.008	1400.584	826.273	1636.588	–	–	–	–
nondet.shallow.32	7644.378	6136.594	13428.044	2150.734	–	28795.875	4258.844	4323.414	3091.523	3155.597	–	–	7358.375	2467.086	6916.869	2040.105
nondet.deep.32	17176.062	8364.722	23378.85	11022.006	–	56373.475	4416.194	9934.269	8049.166	8068.637	–	–	82937.85	6978.472	82501.7	7469.216

Benchmark results compiled with -O0: mean time (% of fastest)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective-naive	mp	mp.safe	mtl	handrolled	mtl.listt	mtl-logict	handrolled.listt	handrolled.logict
countdown.shallow.10000	156.1%	100.0%	340.4%	332.0%	140.2%	170.6%	313.4%	320.3%	165.9%	435.7%	128.9%	109.9%	–	–	–	–
countdown.deep.10000	399.8%	133.2%	385.2%	1462.1%	100.0%	119.3%	227.7%	447.8%	196.8%	709.2%	699.1%	805.0%	–	–	–	–
catch.shallow.10000	482.3%	135.8%	1243.4%	1507.6%	947.1%	166744.2%	416.2%	472.4%	724.4%	726.4%	114.2%	100.0%	–	–	–	–
catch.deep.10000	2568.5%	100.0%	2218.6%	5031.5%	679.0%	123206.7%	330.1%	941.3%	520.5%	519.8%	1255.9%	1280.5%	–	–	–	–
local.shallow.10000	1342.5%	200.1%	804.5%	431.4%	319.2%	115250.1%	191.1%	196.5%	423.1%	421.4%	106.7%	100.0%	–	–	–	–
local.deep.10000	2143.0%	109.1%	897.3%	9644.4%	172.2%	64847.6%	100.0%	259.5%	222.8%	222.7%	131.4%	260.2%	–	–	–	–
nondet.shallow.32	374.7%	300.8%	658.2%	105.4%	–	1411.5%	208.8%	211.9%	151.5%	154.7%	–	–	360.7%	120.9%	339.0%	100.0%
nondet.deep.32	388.9%	189.4%	529.4%	249.6%	–	1276.5%	100.0%	225.0%	182.3%	182.7%	–	–	1878.0%	158.0%	1868.2%	169.1%

Benchmark results compiled with -O0: allocated bytes

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective-naive	mp	mp.safe	mtl	handrolled	mtl.listt	mtl-logict	handrolled.listt	handrolled.logict
countdown.shallow.10000	10640928	6080468	17281848	11920934	5202064	6268896	10801406	11601190	7041752	19361695	4080256	2880380	—	—	—	—
countdown.deep.10000	38643704	14881331	41284510	63365122	5218152	6269549	11213083	15371568	9444743	48965510	30882774	30882810	—	—	—	—
catch.shallow.10000	7913950	2571779	22351478	13075273	12013478	415979526	5433985	6393799	8161791	8161790	1673341	1593381	—	—	—	—
catch.deep.10000	33116268	2572595	57554474	64806042	12032261	416786958	5845704	10164133	8165086	8164582	16926365	18046439	—	—	—	—
local.shallow.10000	14981800	5771832	14008535	7688302	6726751	411513995	4008464	4648320	7193838	7193838	2327805	2327860	—	—	—	—
local.deep.10000	34184815	5772648	31611392	28490607	6742816	412366598	4020157	6818543	7197133	7196567	5048102	9048607	—	—	—	—
nondet.shallow.32	53827057	34532092	81643303	18238376	—	131920487	24814448	29682927	30881957	30881958	—	—	74123029	17396949	69116564	12390830
nondet.deep.32	137695395	64288343	200681211	63405389	—	403887054	26178470	42408046	80929142	80928645	—	—	543610171	43632080	539106548	39124497

Benchmark results compiled with -O0: allocated bytes (% of least allocated)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective-naive	mp	mp.safe	mtl	handrolled	mtl.listt	mtl-logict	handrolled.listt	handrolled.logict
countdown.shallow.10000	369.4%	211.1%	600.0%	413.9%	180.6%	217.6%	375.0%	402.8%	244.5%	672.2%	141.7%	100.0%	—	—	—	—
countdown.deep.10000	740.6%	285.2%	791.2%	1214.3%	100.0%	120.1%	214.9%	294.6%	181.0%	938.4%	591.8%	591.8%	—	—	—	—
catch.shallow.10000	496.7%	161.4%	1402.8%	820.6%	754.0%	26106.7%	341.0%	401.3%	512.2%	512.2%	105.0%	100.0%	—	—	—	—
catch.deep.10000	1287.3%	100.0%	2237.2%	2519.1%	467.7%	16201.0%	227.2%	395.1%	317.4%	317.4%	657.9%	701.5%	—	—	—	—
local.shallow.10000	643.6%	248.0%	601.8%	330.3%	289.0%	17678.2%	172.2%	199.7%	309.0%	309.0%	100.0%	100.0%	—	—	—	—
local.deep.10000	850.3%	143.6%	786.3%	708.7%	167.7%	10257.5%	100.0%	169.6%	179.0%	179.0%	125.6%	225.1%	—	—	—	—
nondet.shallow.32	434.4%	278.7%	658.9%	147.2%	—	1064.7%	200.3%	239.6%	249.2%	249.2%	—	—	598.2%	140.4%	557.8%	100.0%
nondet.deep.32	526.0%	245.6%	766.6%	242.2%	—	1542.8%	100.0%	162.0%	309.1%	309.1%	—	—	2076.6%	166.7%	2059.4%	149.5%